

★ WEEK 11 · PART 1 · PRACTICE

Brownian Motion and SDEs, Worked by Hand

The Volatility Drag, and Why Risk Grows With Root-T

Ito's correction and Brownian scaling, made concrete

Part 1's continuous-time ideas, with the arithmetic shown. We apply Ito's lemma to see why a stock's median falls short of its mean, then use Brownian scaling to compute how volatility and VaR grow with the square root of time.

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PART 1 · APPLICATIONS

Two Continuous-Time Calculations

Case A: geometric Brownian motion and Ito's lemma - the drift correction that splits the mean from the median.

Case B: Brownian variance scaling - the square-root-of-time rule for volatility and VaR. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

Brownian motion is the atom of randomness

Its variance grows with time, so its spread grows with the square root of time. That single fact drives risk scaling.

2

An SDE is drift plus diffusion

Geometric Brownian motion, $dS = \mu S dt + \sigma S dW$, is the standard asset-price model - drift steers, diffusion shakes.

3

Ito's lemma adds a correction

Differentiating a function of a random process leaves an extra half-sigma-squared term. On the log price, it's the volatility drag.

Everything here is a calculator exercise: one drift and vol, one daily volatility, two horizons.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from an SDE or a volatility.
- Apply Ito or the root-T rule by hand.
- Then read the financial consequence.

What you should watch

- How the mean and median part ways.
- How volatility scales with root-T.
- How naive linear scaling overstates risk.

The Volatility Drag

Why the median trails the mean

A high-growth stock has a 10% drift but a hefty 40% volatility. Its price follows geometric Brownian motion. Applying Ito's lemma to the log price reveals a correction - the typical outcome grows not at 10% but at 10% minus half the variance. We compute the gap between the mean and the median by hand

The inputs

- Drift $\mu = 10\%$ per year.
- Volatility $\sigma = 40\%$ per year.
- Start price 100.

What we'll do by hand

- Apply Ito to get the log-price drift.
- Compute the expected (mean) price.
- Compute the median, and the drag.

GBM and Ito on the Log Price

Where the correction comes from

GBM: $dS = \mu S dt + \sigma S dW$ (drift steers, diffusion shakes)

Ito on $\log S$: $d(\log S) = \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma dW$

The price SDE

- Drift and diffusion both scale with S .
- That's what makes returns, not prices, stable.
- The standard model for an asset price.

The log transform

- Ito's lemma adds the -half-sigma-squared term.
- Ordinary calculus would miss it.
- It's the footprint of the roughness.

The Drift Correction

Ten percent becomes two

$$\text{expected log-return} = \mu - \frac{1}{2}\sigma^2 = 0.10 - \frac{1}{2}(0.40)^2 = 0.10 - 0.08 = \mathbf{0.02} \text{ (the Ito correction)}$$

The calculation

- Half of sigma-squared is half of 0.16.
- That's an 8% drag per year.
- Subtracted from the 10% drift.

The result

- The log price drifts at only 2% per year.
- This is the growth a typical path sees.
- Not the headline 10%.

Mean vs Median Price

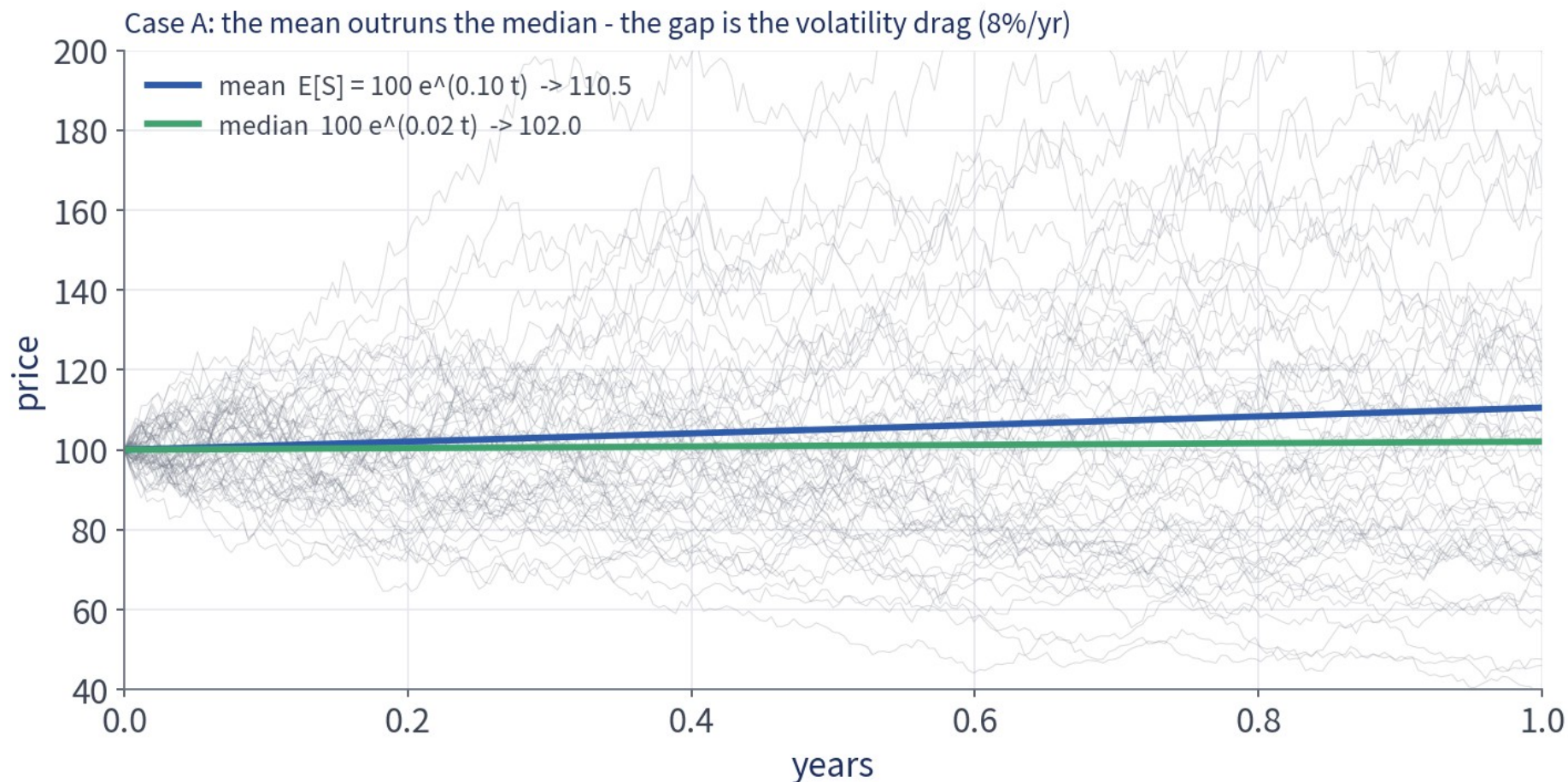
The same stock, two very different numbers

$$\begin{aligned}\text{mean: } \mathbb{E}[S_T] &= S_0 e^{\mu T} = 100 e^{0.10} = \mathbf{110.5} \\ \text{median: } S_0 e^{(\mu - \frac{1}{2}\sigma^2)T} &= 100 e^{0.02} = \mathbf{102.0}\end{aligned}$$

The average price after a year is 110.5, but the typical (median) price is only 102.0. The 8.5-point gap is the volatility drag eating into compound growth.

The Mean Outruns the Median

Simulated GBM paths fan out, but skew: a few big winners pull the mean (blue) up to 110.5, while the typical path - the median (green) - reaches only 102.0. The gap is the volatility drag, and it widens with volatility and time



Why Volatility Costs Growth

The drag is real money

The trap

- Quoting the average return as what you'll get.
- Ignoring that volatility eats compound growth.
- The higher the vol, the bigger the gap.

The reality

- A typical investor sees the median, not the mean.
- Half of sigma-squared is lost to the drag.
- Lower-vol compounding can beat higher-vol drift.

Risk Grows With Root-T

Brownian scaling, applied to VaR

Brownian motion's variance grows with time, so its standard deviation grows with the square root of time. That means a 10-day risk is not ten times a 1-day risk - it's about three times. We compute the 10-day volatility and VaR from a daily figure, and see why linear scaling badly overstates risk

The input

- Daily volatility of 1.5%.
- A 99% confidence VaR.
- Horizons of 1 and 10 days.

What we'll do by hand

- Scale volatility by root-10.
- Compute the 1-day and 10-day VaR.
- Compare to the naive linear answer.

Variance Scales With Time

So volatility scales with root-T

$$\text{Var}(W_t) = t \Rightarrow \text{vol over } T = \sigma\sqrt{T} \quad (\text{risk grows with } \sqrt{T}, \text{ not } T)$$

The property

- The variance of Brownian motion is the time.
- So its standard deviation is root-time.
- Independent increments add variances, not sds.

The consequence

- Volatility over T days is sigma times root-T.
- Risk grows sub-linearly with the horizon.
- The basis of the square-root-of-time rule.

The 10-Day Volatility

From daily to ten days

$$\sigma_{10} = \sigma_{\text{daily}} \sqrt{10} = 1.5\% \times 3.162 = \mathbf{4.74\%} \quad (\text{not } 1.5\% \times 10 = 15\%)$$

The calculation

- Multiply the daily 1.5% by root-10.
- Root-10 is about 3.16.
- That gives 4.74%, not 15%.

The intuition

- Ten days of shocks partly cancel.
- Variances add; volatilities don't.
- So the growth is root-10, not 10.

1-Day vs 10-Day VaR

Risk, scaled honestly

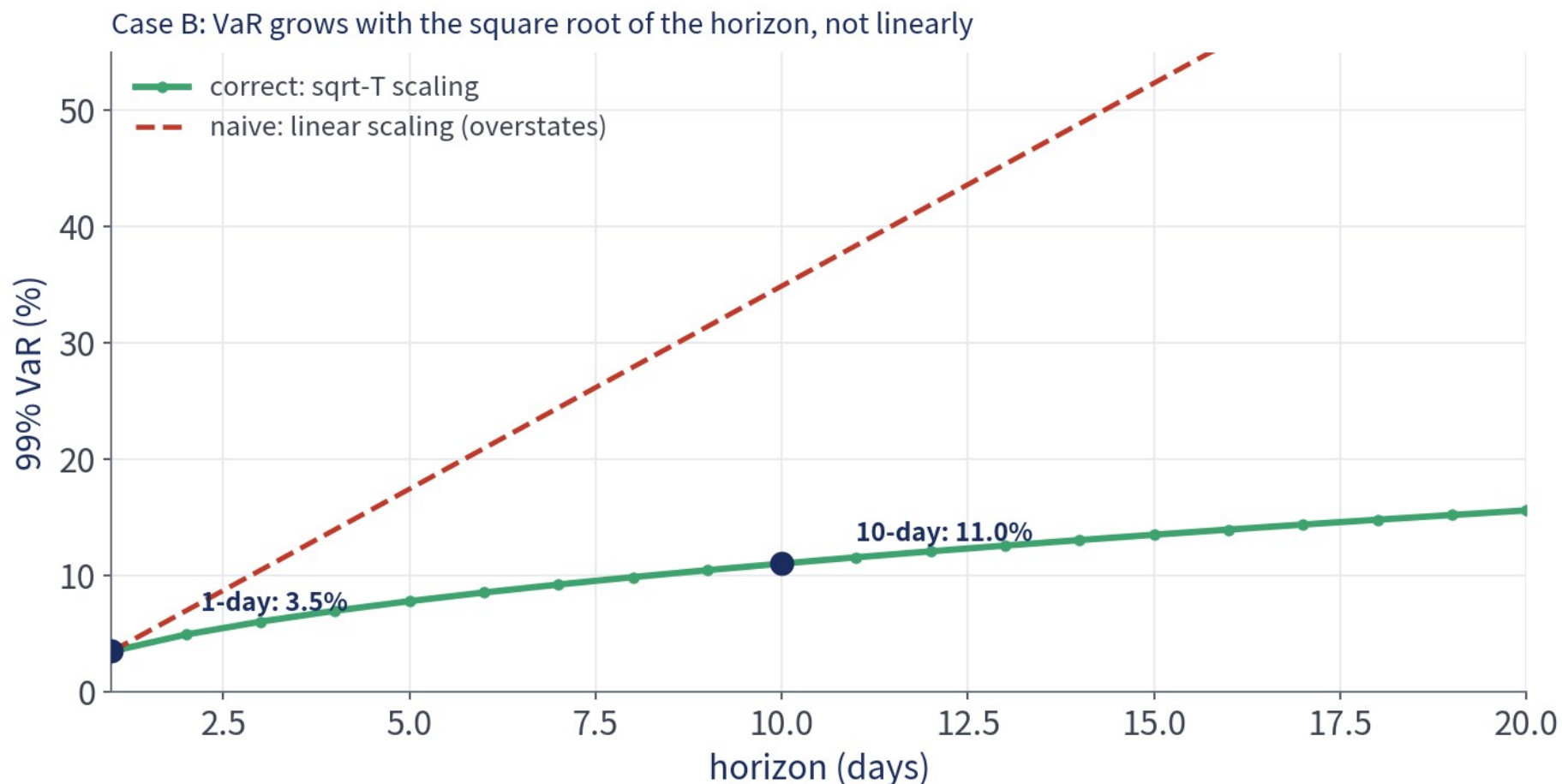
$$\text{1-day 99\% VaR} = 2.326 \times 1.5\% = 3.49\%$$

$$\text{10-day 99\% VaR} = 2.326 \times 1.5\% \times \sqrt{10} = \mathbf{11.0\%} \quad (\text{linear } \times 10 = 34.9\% \text{ is wrong})$$

The 10-day 99% VaR is 11.0%, not the 34.9% you'd get by scaling the 1-day figure linearly. Getting this right is the difference between a sensible and an absurd risk limit.

Root-T, Not Linear

The correct VaR (green) rises with the square root of the horizon, curving and flattening; the naive linear scaling (red) shoots up in a straight line and vastly overstates the risk at longer horizons. At 10 days the difference is 11.0% versus 25.0%



The Square-Root-of-Time Rule

And where it stops working

Why it holds

- Independent shocks add their variances.
- So volatility scales with root-T.
- It's the Basel rule for scaling VaR.

Where it breaks

- If returns are autocorrelated, root-T is off.
- Fat tails and jumps break the Gaussian scaling.
- It's an approximation, not a law.

The Continuous-Time Toolkit

Drift, diffusion, and a correction

Both cases flow from the same continuous-time picture. An SDE splits change into drift and diffusion; Brownian variance grows with time, giving root-T scaling; and Ito's lemma adds the half-sigma-squared correction whenever you transform the process. Get those three, and continuous-time finance opens up

The pieces

- Drift steers; diffusion shakes.
- Variance grows with time; vol with root-T.
- Ito adds the extra half-sigma-squared.

Why they matter

- The drag explains median vs mean.
- Root-T scales risk across horizons.
- Ito is the key to option pricing next.

Reproduce the Drag and the Scaling

Direct Claude Code to check your arithmetic

GOAL Verify the volatility drag and the root-T VaR, then simulate to confirm.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - the drag:

- * from $\mu=0.10$, $\sigma=0.40$, $S_0=100$, print the log-drift $\mu-0.5\sigma^2$, the mean, and the median;
- * simulate GBM paths and check the mean and median.

CASE B - root-T:

- * from daily vol 1.5%, print the 10-day vol, the 1-day and 10-day 99% VaR, and the linear (x10) figure for contrast.

Match every printed number to the slides.

Check against the slides

- log-drift 0.02.
- mean 110.5, median 102.0.
- 10-day vol 4.74%.
- 10-day VaR 11.0% (linear 34.9%).

Deliverable: `part1_practice.py` whose printed numbers match the drag and the root-T scaling.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What is the volatility drag?

**Half sigma-squared,
lost from compound
growth**

Q2 Why does the median trail the mean?

**Ito's correction on
the log price**

Q3 How does VaR scale with the horizon?

**With root-T, not
linearly**

All three come from Ito on the log price and Brownian variance scaling.

Three Pitfalls to Avoid

Where continuous time trips people

Common mistakes

- "The average return is what I'll compound at."
- "Differentiate log price like ordinary calculus."
- "Scale 1-day VaR to 10 days by multiplying by 10."

The reality

- The median grows at $\mu - \frac{1}{2}\sigma^2$.
- Ito adds the half-sigma-squared term - don't drop it.
- Scale by $\sqrt{10}$; linear scaling badly overstates risk.

Two Calculations to Remember

Part 1 applications in one screen

1

The volatility drag splits mean from median

Ito's lemma on the log price turns a 10% drift into a 2% typical growth rate. The mean price reaches 110.5 but the median only 102.0 - the 8%-per-year drag is half of sigma-squared, lost to volatility.

2

Risk grows with the square root of time

Brownian variance scales with time, so the 10-day volatility is 1.5% times root-10 = 4.74%, and the 10-day 99% VaR is 11.0% - not the 34.9% linear scaling wrongly gives. Variances add, not volatilities.

3

Drift, diffusion, and Ito's correction

An SDE separates trend from noise, Brownian motion scales with root-T, and Ito adds the half-sigma-squared term on any transform. These three unlock continuous-time finance - and option pricing ahead.

Next — Part 2 practice: the Ornstein-Uhlenbeck process, mean reversion in continuous time.